Abstract

Operational air quality forecasts are becoming increasingly relevant for public health and environmental decision-making. In Chile, such decision-making is currently supported by SINCA (National Air Quality Information System), which is based on an air quality monitoring network and meteorological forecasts.

Operational air quality prediction in complex urban regions represents a scientific challenge due to the strong dependence of pollutant concentrations on meteorological conditions, topography, and the temporal variability of emissions. In Central Chile, major air pollution episodes are modulated by atmospheric stability processes, coastal ventilation, and topographic confinement, requiring forecasting systems capable of accurately representing the spatial and temporal evolution of atmospheric pollutants.

In this study, we evaluate the capability of a coupled WRF–CHIMERE system to reproduce and forecast the spatiotemporal variability of atmospheric pollutants over Central Chile. To this end, multiple meteorological and chemical simulations were performed using different physical parameterizations, spatial resolutions, and emission inventories. Model performance was assessed through statistical metrics and comparisons against surface observations. The analysis considered both the temporal representation of critical pollution episodes and the spatial distribution of pollutant concentrations under different meteorological regimes.

Introduction

General Objective

To develop and evaluate an operational air quality forecasting system based on the coupling of WRF and CHIMERE for Central Chile, with emphasis on its ability to reproduce and anticipate the spatial and temporal variability of atmospheric pollutants.

Specific Objectives

1. Evaluate the meteorological performance of the WRF model using multiple physical parameterizations and spatial resolutions, considering key variables governing pollutant transport and dispersion, including wind,temperature, atmospheric stability and planetary boundary layer height, in order to select the most appropriate configurations for its coupling with the CHIMERE transport model in order to establish an operational air quality forecast system over Central Chile.

2. Assess the capability of the WRF–CHIMERE system to represent the temporal and spatial variability of atmospheric pollutants (PM25, PM10, CO, and NOx), including the diurnal cycle, high-concentration episodes, temporal persistence, phase errors, forecast window evolution, topographic effects, accumulation and ventilation zones and regional transport patterns.

Methodology

Meteorological System Configuration

The WRF meteorological model was implemented using multiple physical parameterizations and spatial resolutions designed to adequately represent the meteorological conditions of Central Chile, which are characterized by complex topography, land-sea breezes, and seasonal synoptic variability. In particular, the Santiago basin is characterized for its prevailing southwestern summer winds, with a higher frequency of calm, stable days in winter leading to high pollution events.

A forecasting framework was adopted consisting of a 10-day spin-up period followed by ten consecutive daily forecasts during the 2021 winter season. Each of these daily forecasts have a length of 144 hours, for which we derive 4 lead times to consider in the analysis, as represented in the scheme of the Figure 1.

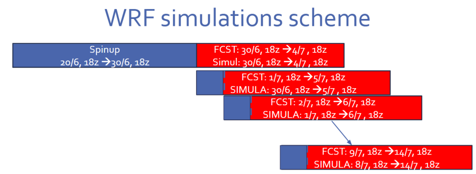

Figure 1: Scheme

For WRF, we tested six physical parameterizations, named by acronyms based on its microphysics, radiation, planetary boundary layer and surface physics schemes as it follows:

- WDMS: WSM-3 microphysics, RRTM/Dudhia radiation, MYNN2.5 PBL, SLAB thermal-diffusion *citation*
- WCMN: CAM longwave radiation, Noah LSM. *citation*
- TDMN: Thompson graupel microphysics scheme, NoahMP surface physics *citation*
- TCYN: YSU PBL scheme *citation*
- WDYN *citation*
- TDYS *citation*

These configurations present unique combinations of microphysics, radiation, PBL and surface physics schemes, among others. Parametrizations are summarized in Table 1

Parameter	WDMS	WCMN	TDMN	TCYN	WDYN	TDYS
mp_physics	3	3	8	8	3	8
ra_lw_physics	1	3	1	3	1	1
ra_sw_physics	1	1	1	3	1	1
radt	15	5	15	5	30	15
sf_sfclay_physics	5	5	1	1	1	1
sf_surface_physics	1	2	4	4	2	1
bl_pbl_physics	5	5	5	1	1	1
bldt	0	0	0	0	0	0
mp_zero_out						2
mp_zero_out_thresh						1.e-8
cu_physics	5	5	5	5	1	2
cudt	5	5	0	5	5	0
cugd_avedx						1
isfflx	1		1	1	1	1
ifsnow	0		0		1	0
icloud	1	1	1	1	1	1
surface_input_source	1		1	1	1	1
num_soil_layers	5		4	5	4	4
topo_wind				1		
slope_rad					1	
topo_shading					1	
maxiens						1
maxens						3
maxens2						3
maxens3						16
ensdim						144
cu_rad_feedback						.true.

Table 1: Physical parametrizations considered for coupling with CHIMERE.

WRF outputs were evaluated using the observational network from the Chilean Meteorological Directorate (DMC) weather stations, AMDAR aircraft profiles from the Pudahuel airport, radiosonde measurements, and SINCA air quality monitoring stations, as shown in Figure 1.

The forecast skill evaluation of the tested WRF configurations was based on traditional statistical metrics, consisting on:

- Normalized Root Mean Square Error (NRMSE)
- Mean Bias Error (MBE)
- Pearson correlation coefficient
- Standard deviation ratio

These metrics were applied across different forecast lead times and spatial resolutions.

Subsequently, and to simplify the decision on which configuration is better suited to connect with the air quality forecasting system, a **multiparametric ranking methodology** was implemented to identify and efficiently communicate the results.

The ranking methodology was designed to provide an objective and comprehensive evaluation of the different configurations of the WRF model by integrating their performance into monitoring stations, vertical atmospheric layers, and forecast settings. The procedure consists of four sequential stages, as illustrated in Figure 1.b.

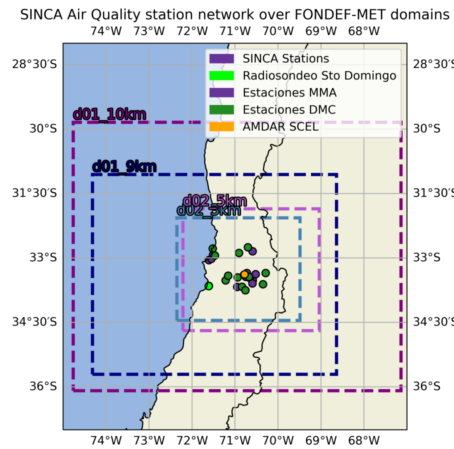

Figure 2: WRF domain configuration and observational network used to evaluate meteorological simulations in Central Chile, including DMC surface stations (dark green), AMDAR profiles (orange), radiosonde observations (located in Santo Domingo, OMM code 85586), and SINCA monitoring sites (purple).

First, time series are extracted from each WRF simulation for every monitoring station and vertical model level. Surface variables are evaluated using station-specific time series, while upper-air variables are analyzed independently at each WRF vertical level. This organization preserves the spatial and vertical variability of the model performance and allows the evaluation to distinguish between near-surface and free-tropospheric behavior. The extracted series are further grouped according to the model configuration, horizontal resolution, and forecast lead time.

In the second stage, statistical performance metrics are computed for every station and vertical level. Four complementary metrics are considered: the Normalized Root Mean Square Error (NRMSE), which quantifies the magnitude of the prediction error; the Pearson correlation coefficient (R), which measures the temporal agreement between simulations and observations; the Mean Bias Error (MBE), which evaluates systematic overestimation or underestimation; and the Standard Deviation Ratio (SD Ratio), defined as the ratio between the simulated and observed standard deviations, which assesses the model's ability to reproduce the observed variability. Together, these metrics provide a balanced description of both accuracy and variability.

The third stage consists of ranking the WRF configurations according to their performance for each metric. A weighted scoring system is then applied to combine the individual rankings into a single performance score. The Normalized Root Mean Square Error (NRMSE) receives a weight of 40%, reflecting its importance as the primary measure of overall forecast accuracy. The Pearson correlation coefficient contributes 30%, emphasizing the capability of the model to reproduce temporal variability. The Mean Bias Error accounts for 20%, penalizing systematic deviations from observations, while the Standard Deviation Ratio contributes the remaining 10%, rewarding configurations that accurately reproduce the observed variance.

Finally, the integrated scores are aggregated across the spatial and vertical domains using a second weighting scheme that reflects the relative importance of different atmospheric layers. Surface performance contributes 40% of the final score, the lower troposphere contributes 45%, and the upper troposphere contributes 15%.

This weighting prioritizes the atmospheric layers that are most relevant for air quality applications while still accounting for the model's performance throughout the atmospheric column.

This hierarchical methodology yields a single objective ranking for each WRF configuration by combining multiple statistical indicators with spatial and vertical performance. The approach minimizes the influence of any individual metric while ensuring that the final ranking reflects both forecast accuracy and the model's ability to reproduce the temporal and vertical structure of the atmosphere.

Chemical Modeling System

The CHIMERE model was coupled with WRF using the two highest-ranked meteorological configurations identified through the ranking analysis.

The system was configured to forecast surface concentrations of PM25, PM10, O3, CO, and NO2 over Central Chile. Simulations were performed using different combinations of:

- Emission inventories (CAMS and INEMA)
- Chemical boundary conditions

The objective was to evaluate system sensitivity and identify the main sources of uncertainty associated with chemical forecasting.

Temporal Forecast Evaluation

Temporal performance was evaluated by comparing simulations against observations from the SINCA monitoring network using hourly time series from representative stations within the study domain.

The analysis focused on:

- Reproduction of the diurnal cycle
- Representation of concentration peaks
- Phase and magnitude errors
- Episode persistence
- Performance degradation with increasing forecast horizon

Additionally, critical pollution episodes were examined to assess the operational capability of the system to anticipate poor air quality events.

Spatial Evaluation

The ability of the system to reproduce the spatial variability of atmospheric pollutants was evaluated through:

- Spatial comparisons between urban and rural monitoring stations
- Evaluation of regional transport and ventilation patterns

Results and discussion

WRF configuration ranking results

In Table 2, we present the results for wind speed and direction between the six WRF configuration, given their more direct involvement with pollutant dispersion. For each configuration, individual cell scores show the relative performance of the configuration compared to the others, for the same resolution and lead time.

Weighted rankings for 10-meter wind speed

WDMS

	24	48	72	96
10 km	3.47	3.35	3.52	3.70
9 km	4.24	3.71	3.77	4.27
5 km	3.67	3.34	3.68	3.66
3 km	3.95	4.08	3.68	4.12
1 km	3.93	3.73	3.39	4.24

TCYN

	24	48	72	96
10 km	2.65	2.54	2.78	2.12
9 km	3.01	2.78	2.61	1.96
5 km	2.48	2.27	2.50	1.83
3 km	2.88	2.49	2.52	1.99
1 km	2.82	2.63	2.38	1.85

WCMN

	24	48	72	96
10 km	4.44	4.06	3.40	3.99
9 km	3.16	3.88	3.58	3.87
5 km	4.50	4.72	3.95	4.51
3 km	3.73	3.54	3.97	3.96
1 km	4.01	3.96	4.36	4.05

WDYN

	24	48	72	96
10 km	4.23	4.40	4.30	4.18
9 km	3.88	3.99	4.58	3.79
5 km	4.34	4.45	3.84	3.83
3 km	3.62	4.06	4.63	3.94
1 km	3.56	4.14	4.40	4.02

TDMN

	24	48	72	96
10 km	3.05	2.96	2.94	3.52
9 km	3.15	2.85	2.90	3.21
5 km	2.57	2.70	2.93	3.39
3 km	3.24	2.98	2.75	3.10
1 km	3.19	3.03	2.89	3.05

TDYS

	24	48	72	96
10 km	3.15	3.70	4.06	3.50
9 km	3.56	3.79	3.55	3.91
5 km	3.44	3.51	4.09	3.79
3 km	3.59	3.84	3.44	3.89
1 km	3.50	3.51	3.58	3.79

Weighted rankings for Wind direction

WDMS

	24	48	72	96
10 km	3.58	2.48	2.68	2.68
9 km	4.07	3.65	4.00	2.72
5 km	3.98	3.29	2.85	3.46
3 km	4.31	3.71	3.74	3.30
1 km	4.28	3.96	4.00	3.31

TCYN

	24	48	72	96
10 km	3.70	3.11	4.28	4.13
9 km	3.03	3.85	2.92	3.01
5 km	3.00	3.00	3.43	4.17
3 km	3.11	3.46	3.26	2.73
1 km	3.17	2.78	3.16	2.93

WCMN

	24	48	72	96
10 km	2.91	4.02	3.04	3.15
9 km	3.70	4.02	3.57	3.93
5 km	4.20	4.28	3.81	3.14
3 km	3.61	4.30	3.62	4.14
1 km	3.90	4.36	3.59	4.30

WDYN

	24	48	72	96
10 km	4.02	4.35	3.68	3.30
9 km	3.43	3.56	3.29	3.18
5 km	3.77	4.12	3.82	2.97
3 km	2.95	3.52	2.98	3.09
1 km	2.77	3.90	2.99	3.48

TDMN

	24	48	72	96
10 km	3.56	3.64	3.75	3.58
9 km	2.91	2.44	2.83	3.72
5 km	2.97	3.08	3.43	3.01
3 km	2.39	1.86	2.78	3.52
1 km	2.64	2.06	2.47	3.08

TDYS

	24	48	72	96
10 km	3.03	3.19	3.16	3.73
9 km	3.66	3.28	3.96	4.03
5 km	2.86	3.02	3.25	3.82
3 km	4.42	3.94	4.22	3.79
1 km	4.05	3.73	4.37	3.48

Table 2. Statistical performance metrics used to evaluate WRF configurations for wind speed (left) and direction (right). These rankings were calculated from evaluating the configurations configurations against observations, including NRMSE, MBE, Pearson correlation coefficient, and standard deviation ratio.

Overall, TYGN dominates in wind speed, arguably in relation to the topographic correction scheme. Nonetheless, this advantage is not as prominent for wind direction, in which TMGN is actually the prevailing configuration at higher resolutions.

The WRF experiments showed that Central Chile's meteorology is strongly controlled by the regional topography, coastal circulation, and seasonal stability conditions. Across the tested configurations, the model reproduced the broad spatial gradients of the observed meteorological fields, with the best-performing setups (VENT, COAST) capturing both the mean state and the main patterns of temporal variability. Differences among configurations were generally modest at the synoptic scale but became more evident in variables directly relevant for air-quality forecasting, especially wind structure and boundary-layer evolution.

The multiparametric ranking procedure allowed us to distinguish configurations that performed similarly under traditional summary metrics but differed in their ability to represent vertical structure and lead-time behavior. The selected top-ranked WRF configurations provided the most balanced performance across surface and upper-air variables, making them suitable for coupling with CHIMERE.

The coupled WRF–CHIMERE system successfully reproduced the main temporal evolution of pollutant concentrations in Central Chile, including the diurnal cycle and the occurrence of high-pollution episodes. Skill was highest for pollutants and stations most directly influenced by large-scale transport and basin-scale ventilation, while larger discrepancies appeared under strongly stagnant conditions and during episodes governed by local emissions and unresolved sub-grid processes. Forecast quality decreased with lead time, but the system retained useful skill over the operational forecast window.

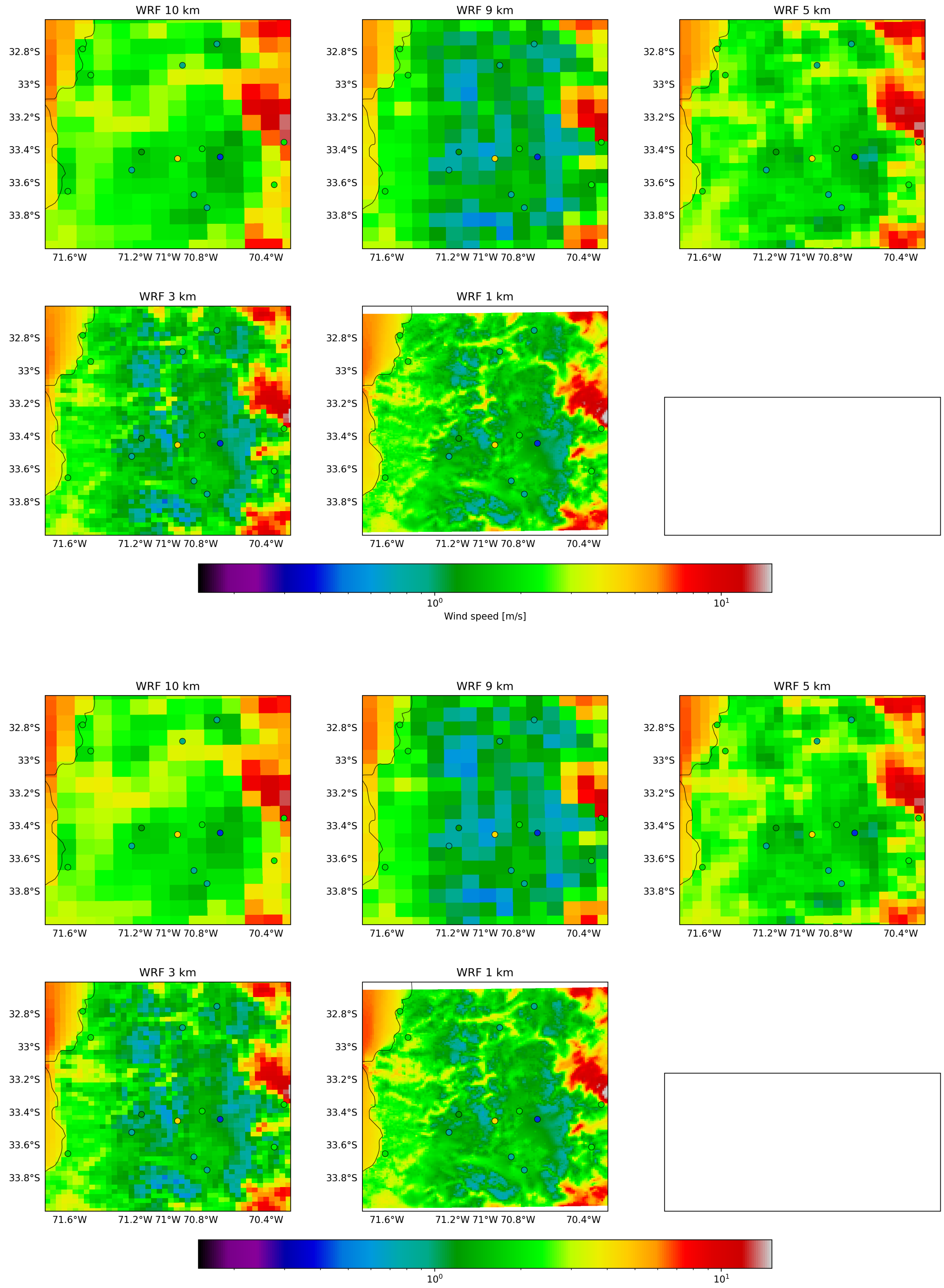

Figure 3 Temporal mean of simulated meteorological variables across Central Chile, compared with station observations for the tested WRF configurations.

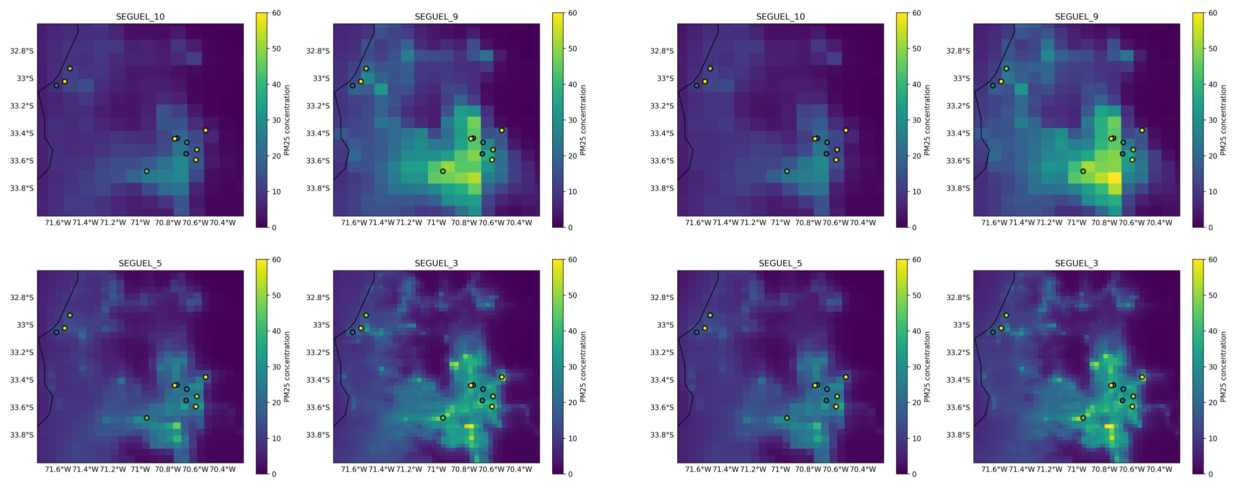

Figure 4 Forecast evaluation of pollutant concentrations from the coupled WRF–CHIMERE system at SINCA stations (scatter plots), showing observed and simulated diurnal cycles and episode evolution for lead times of 24 hours (upper) and 96 hours (lower).

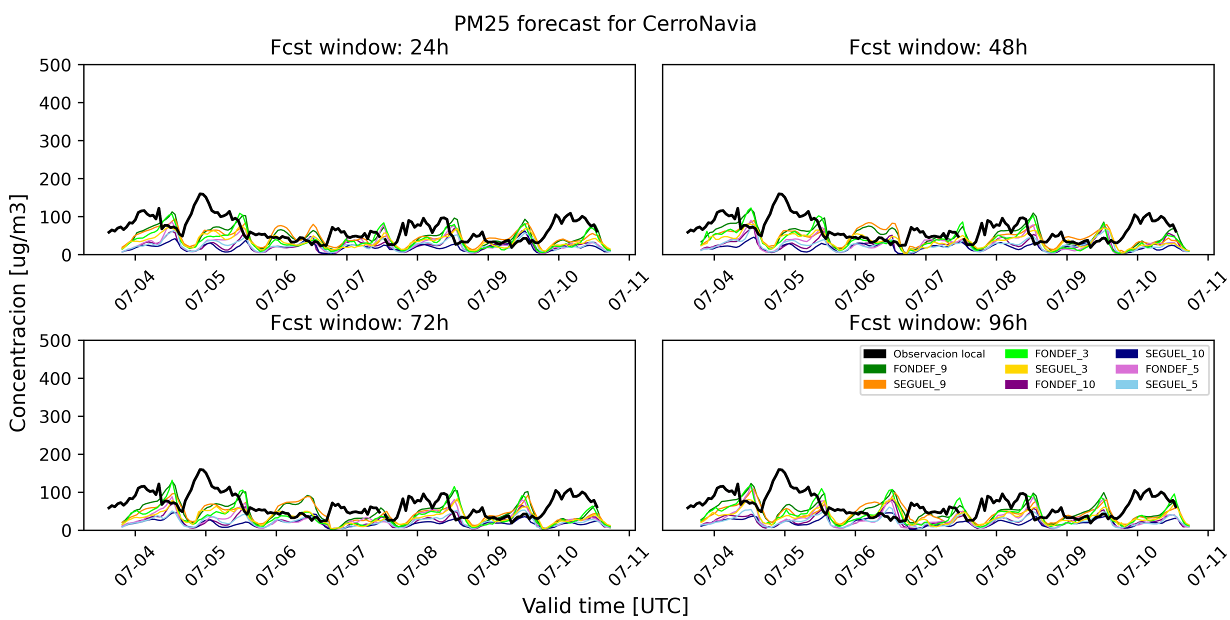

Figure 5 Forecast evaluation of pollutant concentrations from the coupled WRF–CHIMERE system at SINCA stations, showing observed and simulated diurnal cycles and episode evolution.

Spatially, the model captured the contrast between accumulation zones in the Santiago basin and better-ventilated surrounding areas. The simulations also reproduced the regional influence of transport, particularly under wintertime stability conditions when pollutants were more likely to persist and accumulate. Sensitivity experiments using different emissions inventories and boundary conditions showed that model performance was meaningfully affected by these inputs, confirming their importance as major sources of uncertainty in operational forecasting.

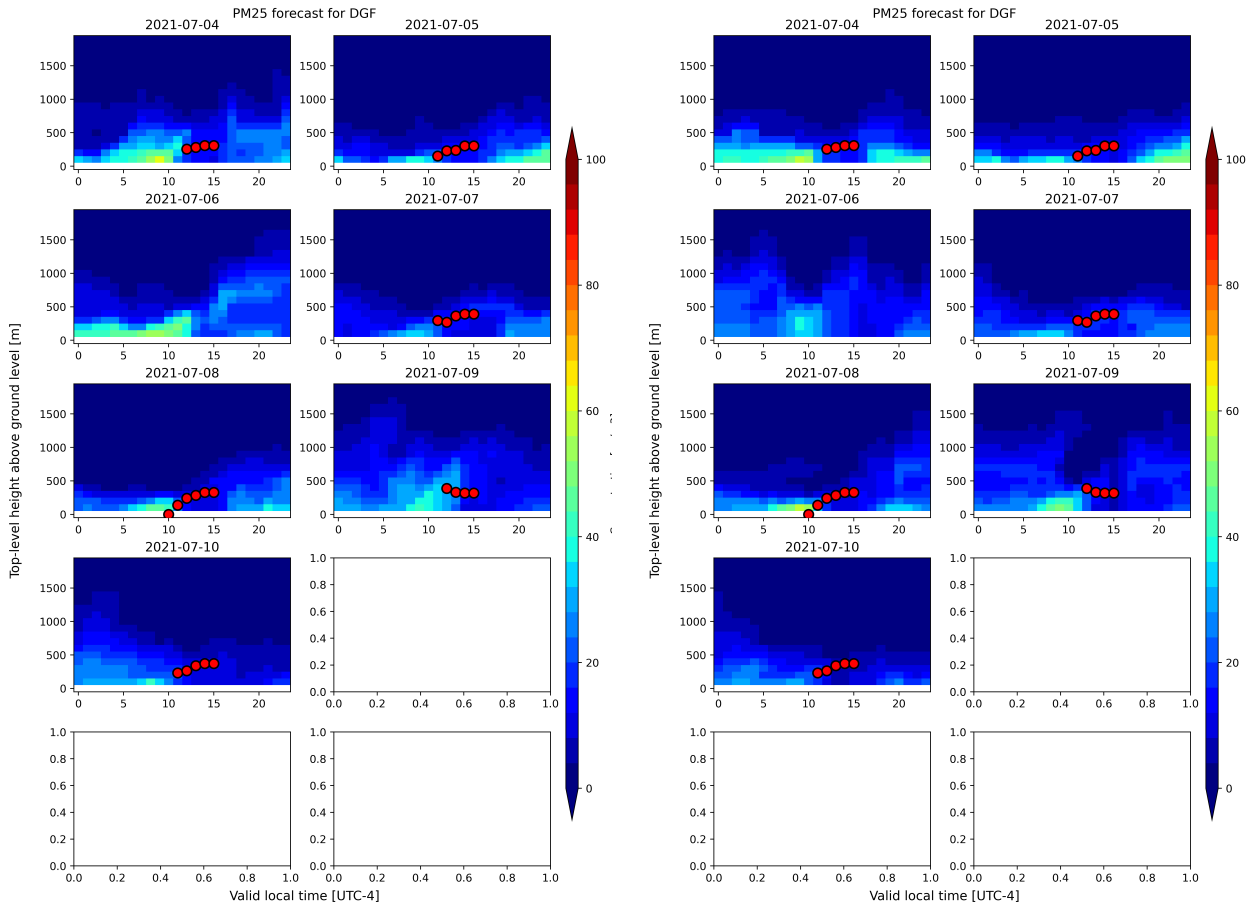

Figure 6 Spatial representation of modeled pollutant concentrations and their temporal variability under different meteorological regimes and emission scenarios.

To evaluate the spatial performance of the meteorological simulations, the temporal mean and temporal standard deviation of the modeled variables were compared with observations at all monitoring stations (Figures XX–XX). While the mean fields characterize the climatological state reproduced by the model, the standard deviation provides information on its ability to represent temporal variability and meteorological fluctuations.

The spatial distribution of the temporal mean indicates that the WRF configurations successfully reproduce the large-scale spatial gradients observed across Central Chile. For most variables, the observed station values closely follow the simulated background field, indicating that the model captures the first-order influence of topography, land–sea contrasts, and regional circulation patterns on the climatological state. Remaining discrepancies are generally localized and tend to occur at stations situated in areas of complex terrain, where unresolved topographic effects and local circulations become increasingly important.

The standard deviation fields provide complementary information regarding the temporal dynamics of the simulations. In general, the model reproduces the geographical distribution of variability reasonably well, indicating that regions exhibiting stronger meteorological fluctuations are consistently identified by the simulations. This agreement suggests that the model captures the dominant synoptic and mesoscale processes controlling temporal variability throughout the study domain.

However, comparison with the observations reveals localized differences in the magnitude of the temporal variability. At several stations the model either underestimates or overestimates the observed standard deviation despite reproducing the mean state adequately. These results indicate that achieving a realistic climatological average does not necessarily imply an accurate representation of temporal fluctuations. Such discrepancies likely reflect limitations in the representation of local processes, including boundary-layer evolution, terrain-induced circulations, coastal breezes, and sub-grid-scale turbulence.

Taken together, the temporal mean and standard deviation demonstrate that the evaluated WRF configurations reproduce both the climatological spatial structure and the principal patterns of meteorological variability with satisfactory skill. The combined analysis also highlights that model performance should not be evaluated solely through mean-state statistics, since deficiencies in temporal variability may propagate into errors in pollutant transport, atmospheric mixing, and the timing of air-quality episodes.

The comparison among WRF configurations reveals only modest differences in both diagnostics, consistent with the ranking analysis presented previously. Although the highest-ranked configurations provide slightly better agreement with observations, the overall spatial patterns remain similar across simulations, suggesting that the selected physical parameterizations primarily influence local-scale behavior rather than the large-scale meteorological structure. This supports the use of the top-ranked configurations for the subsequent coupling with the CHIMERE chemical transport model.

Annex

WRF metrics and ranking

PM25

PBL analysis

Conclusions

Bibliography

1. Lapere, R., Menut, L., Mailler, S., & Huneeus, N. (2021). Seasonal variation in atmospheric pollutants transport in central Chile: Dynamics and consequences. *Atmospheric Chemistry and Physics, 21*(8), 6431–6454. https://doi.org/10.5194/acp-21-6431-2021

2. Saide, P. E., Mena-Carrasco, M., Tolvett, S., Hernandez, P., & Carmichael, G. R. (2016). Air quality forecasting for winter-time PM2.5 episodes occurring in multiple cities in central and southern Chile. *Journal of Geophysical Research: Atmospheres, 121*(1), 558–575. https://doi.org/10.1002/2015JD023949

3. Gao, C., Zhang, X., Xiu, A., Tong, Q., Zhao, H., Zhang, S., Yang, G., Zhang, M., & Xie, S. (2024). Intercomparison of multiple two-way coupled meteorology and air quality models (WRF v4.1.1–CMAQ

v5.3.1, WRF–Chem v4.1.1, and WRF v3.7.1–CHIMERE v2020r1) in eastern China. *Geoscientific Model Development, 17*(6), 2471–2492. https://doi.org/10.5194/gmd-17-2471-2024